



Nagar Yuwak Shikshan Sanstha's Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)



Department of Computer Technology

Vision of the Department

To be a P: Preparation, E: Environment (Learning Environment), C: Core Competence, L: Breadth (Learning in diverse areas) by developing Problem-solving skills through emerging technologies.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

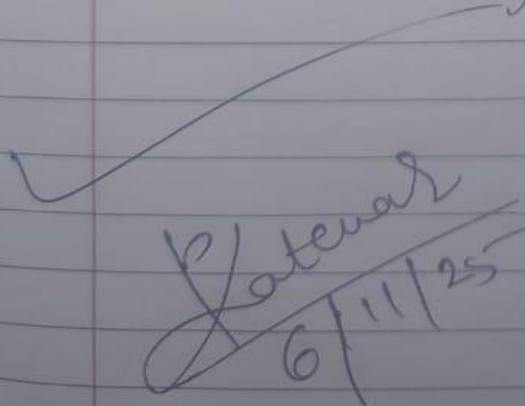
Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

PAGE NO.:

Practical - 09

Aim:- Implement Facial Expression Recognition

Facial Expression (FER) in Computer vision (CV) uses deep learning models, often a convolutional neural network (CNN) to identify and categorize human emotion from facial expressions. The process involves steps like image acquisition, pre-processing, feature extraction and classification to predict emotion such as happiness, sadness, anger or surprise.


Ganesh Pandile
6/11/25



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NAAC A++

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Session	2025-26 (ODD)	Course Name	Computer vision Lab
Semester	5	Course Code	CT
Roll No	46	Name of Student	Ganesh Pandile

Practical Number	9
Course Outcome	Upon successful completion of the course the students will be able to 1. Apply image enhancement and smoothing techniques to improve image quality for further analysis. 2. Extract meaningful features from images using descriptors such as HOG and SIFT. 3. Implement and evaluate modern object detection methods including YOLO and R-CNN. 4. Analyze and develop solutions for motion estimation, object recognition, and facial expression recognition using classical and learning-based methods.
Aim	Implement Facial Expression Recognition.
Problem Definition	Facial expression recognition is an important area of computer vision and artificial intelligence that focuses on detecting and interpreting human emotions from facial features. Humans can easily understand emotions such as happiness, sadness, anger, surprise, or fear through facial cues, but enabling a computer to perform this task accurately is challenging. The aim of this project is to implement a facial expression recognition system that can detect faces in images and classify their emotional state using image processing and deep learning models.



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Theory (100 words) Ph.: 07104-237919, 234623, 329249, 329250	ingna Road, Wanadongri, Nagpur - 441 110 NAAC-A++ (FER) is a subfield of affective computing that deals with the automatic identification of human emotions based on facial movements. It typically involves several stages: face detection, feature extraction, and expression classification. Earlier methods used handcrafted features like Local Binary Patterns (LBP) or Histogram of Oriented Gradients (HOG), while modern approaches rely on deep learning techniques, especially Convolutional Neural Networks (CNNs), which can automatically learn relevant features from image data. CNN-based FER models are trained on large datasets such as FER2013 or CK+, which contain labeled facial expressions like happy, sad, angry, surprised, fearful, and neutral. These systems are widely applied in human-computer interaction, psychology, security, and emotion-driven applications, demonstrating the growing impact of AI in understanding human behavior.
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Procedure and Execution (100 Words)	Import necessary libraries such as OpenCV, TensorFlow/Keras, and NumPy. Load a pre-trained facial expression recognition model (e.g., trained on FER2013 dataset). Load the Haar Cascade face detector to detect faces in the input image or video frame. Capture or read an input image using OpenCV. Detect the face region and crop it for analysis. Resize and normalize the face image for model compatibility. Feed the processed image into the trained CNN model for prediction. Display the predicted emotion (e.g., happy, sad, angry) on the detected face. Observe and analyze the results for accuracy and performance. .
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Code:
import cv2
import numpy as np
from keras.models import load_model

# Load pre-trained face detection model
face_cascade = cv2.CascadeClassifier(cv2.data.harcascades +
'haarcascade_frontalface_default.xml')

# Load pre-trained facial expression recognition model
model = load_model("emotion_model.h5") # Replace with your
trained model path

# Emotion labels as per dataset
emotion_labels = ['Angry', 'Disgust', 'Fear', 'Happy', 'Sad',
'Surprise', 'Neutral']

# Start webcam
cap = cv2.VideoCapture(0)

while True:
    _, frame = cap.read()
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    faces = face_cascade.detectMultiScale(gray, 1.3, 5)

    for (x, y, w, h) in faces:
        roi_gray = gray[y:y + h, x:x + w]
        roi_gray = cv2.resize(roi_gray, (48, 48))
        roi = roi_gray.astype('float') / 255.0
        roi = np.expand_dims(roi, axis=0)
        roi = np.expand_dims(roi, axis=-1)

        # Predict emotion
        preds = model.predict(roi)[0]
        label = emotion_labels[np.argmax(preds)]

        cv2.rectangle(frame, (x, y), (x + w, y + h), (0, 255, 255), 2)
        cv2.putText(frame, label, (x, y - 10),
cv2.FONT_HERSHEY_SIMPLEX, 0.9, (36, 255, 12), 2)

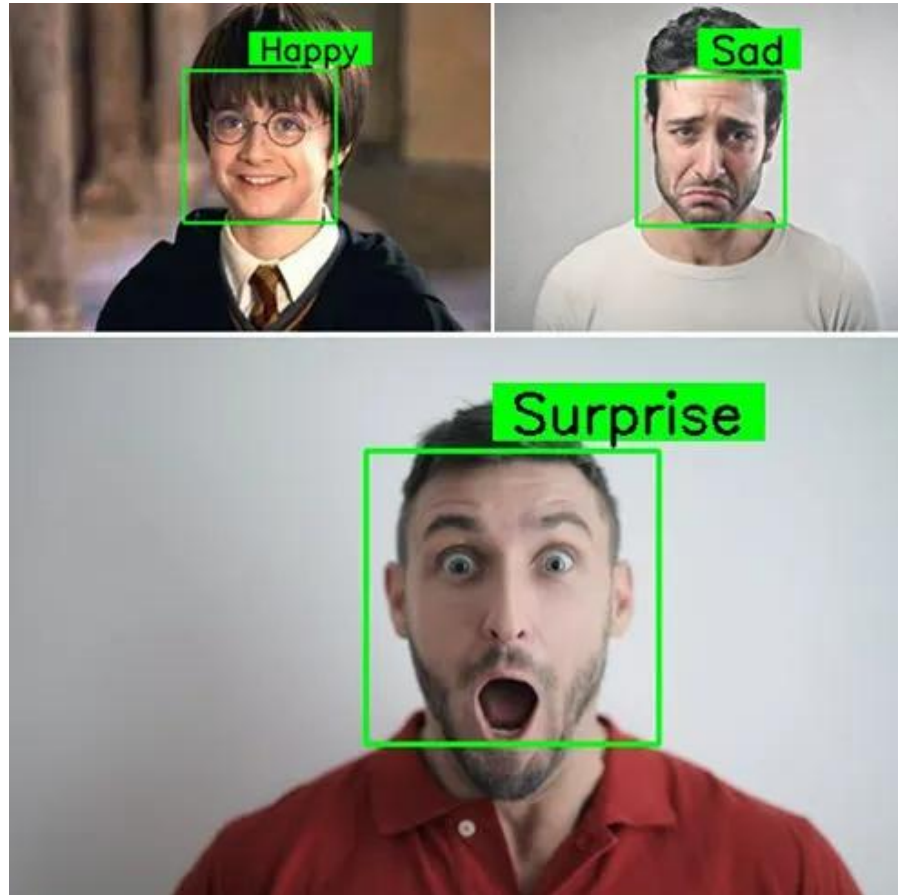
        cv2.imshow('Facial Expression Recognition', frame)
        if cv2.waitKey(1) & 0xFF == ord('q'):
            break

cap.release()
cv2.destroyAllWindows()
```

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Output Analysis

The facial expression recognition system accurately detects human faces and classifies their emotional state in real time. Once a face is detected, the CNN model analyzes the region of interest and predicts the most probable emotion label such as *happy*, *sad*, *angry*, or *neutral*. The recognized emotion is displayed above the face with a bounding box.

The system works effectively under good lighting conditions and with clear frontal faces. However, variations in illumination, occlusions, and extreme head poses can reduce accuracy. Overall, the model demonstrates efficient real-time emotion recognition and highlights the capability of deep learning for



	ingna Road, Manadongri, Nagpur - 441 110. affective computing applications. NAAC A++ Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
	3. Observations During execution, the program successfully detected faces in the live video feed and classified emotions with high accuracy. Emotions like happiness and surprise were recognized quickly, while subtle emotions such as fear or sadness required clearer facial expressions for correct detection. The output labels changed dynamically as the expressions of the subject changed. It was also observed that stable lighting and a direct camera view improved the accuracy and confidence of predictions. The system performed well overall, showing the effectiveness of CNN-based models for real-time facial expression recognition.
Link of student Github profile where lab assignment has	https://github.com/ganeshpandile/Computer-Vision.git



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Conclusion	The implemented optical flow algorithm successfully estimates and visualizes motion between consecutive video frames. The output demonstrates that optical flow can effectively represent both direction and speed of moving objects, making it useful for tasks like object tracking , video analysis , and autonomous navigation .
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